

Question number	Answer	Mark		
3(a)(i)	The only correct answer is B <table><tr><td>contracted</td><td>relaxed</td></tr></table> <i>A is incorrect because the ventricles are relaxed</i> <i>C is incorrect because the atria are contracted and the ventricles are relaxed</i> <i>D is incorrect because the atria are contracted</i>	contracted	relaxed	(1)
contracted	relaxed			

Question number	Answer	Additional guidance	Mark
3(a)(ii)	An explanation that includes the following points: - because the atrioventricular valves have to close (before the ventricles contract) (1) - to prevent backflow of blood into the atria (1)	ACCEPT {bicuspid / mitral} valve and tricuspid valve DO NOT ACCEPT valves close during ventricular systole	(2)

Question number	Answer	Additional guidance	Mark
3(a)(iii)	- cardiac cycle time multiplied by proportion of cycle spent in ventricular systole (1) $3.2 \times 10^2 / 3.23 \times 10^2$ (1)	Example of calculation: $0.86 \times \frac{3}{8} / 0.3225 / 0.32 / 0.323$ Correct answer with no working gains 2 marks	(2)

Question number	Answer	Additional guidance	Mark
3(a)(iv)	63% / 5/8 / 0.63	ACCEPT 0.625 / 62.5% / 5 out of 8	(1)

Question number	Answer	Additional guidance	Mark
3(b)	- heart rate if cardiac cycle lasts 0.86 seconds (1) - heart rate if cardiac cycle last 0.46 seconds (1) - increase in heart rate = 60 / 61 beats per minute (1)	Example of calculation: $60 \div 0.86 = 69.76744186046512$ $60 \div 0.46 = 130.4347826$ Answer in range of 60.2 to 60.7 gains 2 marks CE from calculations of heart rate Correct answer alone gains 3 marks	(3)